

9. (a) A student tested compounds **A**, **B** and **C** to identify them.

Her observations are recorded in the tables below.

| Compound <b>A</b>           | Observation       | Ion present |
|-----------------------------|-------------------|-------------|
| Flame test                  | apple-green flame | barium      |
| Add silver nitrate solution | white precipitate | chloride    |

| Compound <b>B</b>           | Observation       | Ion present |
|-----------------------------|-------------------|-------------|
| Flame test                  | lilac flame       |             |
| Add silver nitrate solution | cream precipitate |             |

| Compound <b>C</b>           | Observation        | Ion present |
|-----------------------------|--------------------|-------------|
| Flame test                  | brick-red flame    |             |
| Add silver nitrate solution | yellow precipitate |             |

- (i) Use your knowledge of the tests for ions to complete the tables for compounds **B** and **C**.

[3]

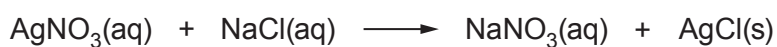
- (ii) Write the chemical formula for compound **A**.

[1]

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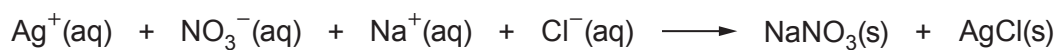
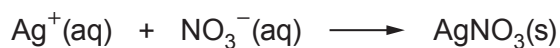
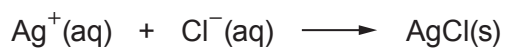
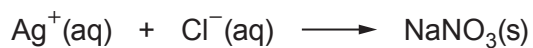


- (b) When silver nitrate solution reacts with sodium chloride solution to produce a white precipitate the following reaction occurs.



Put a tick (✓) in the box which shows the ionic equation for the formation of the precipitate.

[1]

☐☐☐☐☐

- (c) When silver carbonate is heated it breaks down to give silver metal and carbon dioxide and oxygen gas.



Calculate the mass of silver that would be produced from heating 13.8 g of silver carbonate.

[3]

$$M_r(\text{Ag}_2\text{CO}_3) = 276 \quad A_r(\text{Ag}) = 108$$

Mass = ..... g

**END OF PAPER**

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